🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.NOS.1

Division Expressions with Fractions and Whole Numbers

Lesson 6-1 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

EVIDENCE LAB MISSION

The Fraction Vault

You are the lead forensic analyst at the Reyes Detective Agency. A 3-foot strip of evidence tape must be cut into exact 1/4-foot sections before it can be logged — and the vault that holds the case file only opens once you know how many pieces that makes. Interpret fraction division and crack the case.

My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can divide a whole number by a unit fraction and a unit fraction by a whole number by writing the whole number over 1 and using Keep, Change, Flip.

Language Objective: I can explain each step of Keep, Change, Flip using the words reciprocal, divisor, and quotient.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Division of Fractions | **Reciprocal** | **Quotient** | **Dividend** | **Divisor** | **Unit fraction**

👆 *Tap any word to see what it means and a picture.*

1. Dividing asks how many groups of one ______ fit inside another amount.

2. Flipping the numerator and denominator gives the ______.

3. The answer to a division problem is the ______.

4. In 6 ÷ 1/2, the number 6 being divided is the ______.

5. In 6 ÷ 1/2, the number 1/2 we divide by is the ______.

6 A fraction with a numerator of 1, like 1/5, is a ______________ .

Write About the Math

How does interpreting fraction division help the detective?

¿Cómo ayuda interpretar la división de fracciones al detective?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Division of Fractions** — *División de fracciones*
- ☐ **Reciprocal** — *Recíproco*
- ☐ **Quotient** — *Cociente*
- ☐ **Dividend** — *Dividendo*
- ☐ **Divisor** — *Divisor*

Model: *There will be more than 3 sections because each piece is smaller than a whole foot.* — Students reason that dividing by a number less than 1 gives a quotient greater than the dividend, so there are more than 3 sections.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The detective can mark ____ sections because 6 ÷ 1/3 = ____.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Division of Fractions
2. **Notes 2:** Reciprocal
3. **Notes 3:** Quotient
4. **Notes 4:** Dividend
5. **Notes 5:** Divisor
6. **Notes 6:** Unit fraction
7. **Watch for this mistake:** *A common mistake in Interpret Division of Fractions is assuming that dividing always makes the answer smaller. For 4 ÷ 1/3, students reason 'dividing shrinks the number' and compute 4 ÷ 3 ≈ 1.3 instead of asking how many 1/3-foot pieces fit into 4 feet (each whole foot holds 3 thirds, so 4 × 3 = 12). Before you submit, check: if the divisor is a fraction smaller than 1, the quotient must be BIGGER than the dividend, not smaller.*
8. **Try It 1:** A. 5 ÷ 1/6 = 30 slivers — *How many 1/6-inch slivers fit into 5 inches? Put a 1 under the 5, then Keep, Change, Flip: 5/1 × 6/1 = 30 slivers.*
9. **Try It 2:** *— 3/4 ÷ 1/4 asks how many one-fourth pieces make up three-fourths. The fraction bar shows 3/4 already built out of fourth-size pieces, so counting those pieces gives the quotient: 3. Keep, Change, Flip agrees — 3/4 × 4/1 = 12/4 = 3. Neither number is a whole number here, so there is nothing to write over 1.*